

Probability and Random Variable

Module 3

Subject Name: Statistics and
Probability

Subject Code: BAS0303N

B. Tech 3rd Sem



Dr. Jyoti Singh
Department of Mathematics

Module 1: (Statistical Techniques-I)

Introduction: Measures of central tendency: Mean, Median, Mode, Standard deviation, Quartile deviation, Moment, Skewness, Kurtosis.

Module 2: (Statistical Techniques-II)

Curve Fitting, Method of least squares, fitting of straight lines, Fitting of second-degree parabola, Exponential curves, Correlation and Rank correlation, Linear regression, nonlinear regression and multiple linear regression.

Module 3: (Probability and Random Variable)

Random Variable: Definition of a Random Variable, Discrete Random Variable, Continuous Random Variable, Probability mass function, Probability Density Function, Distribution functions.

Multiple Random Variables: Joint density and distribution Function, Properties of Joint Distribution function, Marginal density Functions, Conditional Distribution and Density, Statistical Independence, Central Limit Theorem (Proof not expected).

Module 4: (Expectations and Probability Distribution)

Expectations of single Random Variable, Mean, Variance, Moment Generating Function, Binomial, Poisson, Normal, Exponential distribution.

Module 5: (Hypothesis Tests and Control Charts)

Testing a Hypothesis, Null hypothesis, Alternative hypothesis, Level of significance, Confidence limits, Test of significance of difference of means, Z-test, t-test and Chi-square test, F-test, One way ANOVA.

Statistical Quality Control (SQC), Control Charts, Control Charts for variables (Mean and Range Charts), Control Charts for Variables (p, np and C charts).

- ❖ Data Analysis
- ❖ Artificial intelligence
- ❖ Network and Traffic modeling

Course Objectives

- The objective of this course is to familiarize the students with concepts of Probability and statistical techniques. It aims to equip the students with adequate Knowledge of statistics that will enable them in formulating Problems and solving problems analytically.

The students will learn:

- Understand the concept of correlation, moments, skewness and kurtosis and curve fitting.
- Apply the concept of hypothesis testing and statistical quality control to create control charts.
- Remember the concept of probability to evaluate probability distributions.
- Understand the concept of Mathematical Expectations and Probability Distribution.

CO1: Apply the concept of moments, skewness and kurtosis in relevant field.

CO2: Apply the concept of correlation, regression and curve fitting with real world problems.

CO3: Apply the concept of probability and random variable.

CO4: Apply the concept of Mathematical Expectations and Probability Distribution in real life problems.

CO5: Apply the concept of hypothesis testing and statistical quality control to create control charts.

- Knowledge of Maths 1 B.Tech.
- Knowledge of Maths 2 B.Tech.
- Knowledge of Permutation and Combination.

Module Objectives (CO3)

1. A basic knowledge of random variables.
2. The student is able to reflect developed mathematical methods in probability and random variable.
3. Understand the concept of random variable.
4. To explore the key properties: such as PMF, PDF etc.

Topic Objective(CO3)

The random variable is very much helpful for making prediction. Estimates and predictions form an **important** part of research investigation. With the help of statistical methods, we make estimates for the further analysis.

Definition: Random variable is a real-valued function whose domain is the sample space of a random experiment and codomain is a real number

Symbol : X

Example : An urn contains 5 red and 2 black balls. Two balls are randomly drawn. Let X represents the number of black balls. What are the possible values of X ? Is X a random variable?

Solution:

Let B represents a black ball and R represents a red ball.

The two balls selected can be represented as BB , BR , RB , RR .

Thus, the possible values of X are 0, 1, and 2.

Yes, X is a random variable.

There are two types of Random Variable.

Discrete Random variable: A random variable X , which can take only a finite number of values in an interval of the domain is called discrete random variable.

Example:

- Number of mistakes in a page
- Number of defective items in a lot
- By tossing a coin 4 times, the number of Heads come up then the value of random variables are 0,1,2,3,4.

Continuous Random Variable: which variables can assume any value within interval is called continuous .

Example :

- The heights of a group of individuals,
- Weight of the students in a class etc.

Probability Mass Function / probability distribution(CO3)

Probability Mass Function: A function $P(x)$ or $p(x)$ defined by

$$P(X = x) = p(x) = \begin{cases} p(x_i) \text{ or } p_i & \text{where } x = x_i, i = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

is called Probability function of the **discrete random variable** X . where X assumes values $x_1, x_2, x_3, \dots, x_n$, and $p_1, p_2, \dots, p_n$ are corresponding probabilities. In short, it is called pmf

Properties:

- i. $p(x_i) \geq 0$
- ii. $\sum p(x_i) = 1$

Probability distribution of a Random Variable(CO3)

Example: Is $P(X)$ a Probability distribution ?

X	0	1	2
P(X)	0.4	0.4	0.2

Solution: Yes, it is a Probability distribution because sum of all Probability ($0.4+0.4+0.2= 1$) is 1

Example:2 Find the probability distribution of number of heads in two tosses of a coin.

Solution: When one coin is tossed twice. the sample space is $\{HH,HT,TH,TT\}$
Let X represent the number of heads. X can take the value of 0, 1, or 2.

X	0	1	2
P(X)	1/4	2/4	1/4

Question 1: State which of the following are not the probability distribution of a random variable. Give reasons for your answer.

(i)

X	0	1	2
P (X)	0.4	0.4	0.2

(ii)

X	0	1	2	3	4
P (X)	0.1	0.5	0.2	-0.1	0.3

(iii)

X	-1	0	1
P (X)	0.6	0.1	0.2

(iv)

X	3	2	1	0	-1
P (X)	0.3	0.4	0.1	0.3	0.05

1. Mean of discrete random variables:

$$\text{Mean } \mu = \sum p_i x_i$$

2. Variance of discrete random variables:

$$\text{Variance} = \sum p_i x_i^2 - \mu^2$$

Q1. Five defective bulbs accidentally mixed with twenty good ones. It is not possible to just look at a bulb and tell whether or not it is defective. Find the probability distribution of the number of defective bulbs, if four bulbs are drawn at random from this lot.

Solution: Let X denote the number of defective bulbs in 4 so X can take values 0,1,2,3 and 4.

Number of defective bulbs =5

Number of good bulbs =20

Total number of bulbs =25

$$\begin{aligned} P(X = 0) &= P(\text{no defective}) = P(\text{all 4 good ones}) \\ &= \frac{{}^{20}C_4}{{}^{25}C_4} = \frac{20 \times 19 \times 18 \times 17}{25 \times 24 \times 23 \times 22} = \frac{969}{2530} \end{aligned}$$

$$P(X=1)=P(\text{ one defective and 3 good ones})=\frac{{}^5C_1 \times {}^{20}C_3}{{}^{25}C_4} = \frac{1140}{2530}$$

$$P(X=2)=P(2 \text{ defective and } 2 \text{ good ones})=\frac{{}^5C_2 \times {}^{20}C_2}{{}^{25}C_4} = \frac{380}{2530}$$

$$P(X=3)=P(3 \text{ defective and } 1 \text{ good ones})=\frac{{}^5C_3 \times {}^{20}C_1}{{}^{25}C_4} = \frac{40}{2530}$$

$$P(X=4)=P(\text{all } 4 \text{ defectives})=\frac{{}^5C_4}{{}^{25}C_4} = \frac{1}{2530}$$

Therefore probability distribution of the random variable X is

x	0	1	2	3	4
P(x)	$\frac{9669}{2530}$	$\frac{1140}{2530}$	$\frac{380}{2530}$	$\frac{40}{2530}$	$\frac{1}{2530}$

Q 2. A die is tossed thrice . A success is ‘getting 1 or 6’ on a toss . Find the mean and the variance of the number of success.

Solution: Let X denote the number of success . Clearly X can take the values 0,1,2 or 3.

$$\text{Probability of success} = \frac{2}{6} = \frac{1}{3}$$

$$\text{Probability of failure} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(X=0) = P(\text{no success}) = P(\text{all 3 failures}) = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$$

$$P(X=1) = P(1 \text{ success and 2 failure}) = {}^3C_1 \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{12}{27}$$

$$P(X=2) = P(2 \text{ success and 1 failure}) = {}^3C_2 \times \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{6}{27}$$

$$P(X=3) = P(\text{all 3 successes}) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$$

Therefore the probability distribution of the random variable X is

X	0	1	2	3
P(X)	$\frac{8}{27}$	$\frac{12}{27}$	$\frac{6}{27}$	$\frac{1}{27}$

To find the mean and Variance

$$\begin{aligned} \text{Mean } \mu &= \sum p_i x_i = x_0 p_0 + x_1 p_1 + x_2 p_2 + x_3 p_3 \\ &= 0 \times \frac{8}{27} + 1 \times \frac{12}{27} + 2 \times \frac{6}{27} + 3 \times \frac{1}{27} = 1 \end{aligned}$$

$$\text{Variance} = \sum p_i x_i^2 - \mu^2 = \frac{5}{3} - 1 = \frac{2}{3}$$

Continuous Random Variable: Which variables can assume any value within interval is called continuous.

Example : The heights of a group of individuals, Weight of Students etc.

Probability Density function: It is expressed as

$$P\left(x - \frac{1}{2}dx \leq X \leq x + \frac{1}{2}dx\right) = f(x)dx$$

Properties of probability Density function:

- i. $f(x) \geq 0$ for every x ,
- ii. $\int_{-\infty}^{\infty} f(x)dx = 1$

Q1. If $f(x)$ has probability density cx^2 , $0 < x < 1$, determine c and find the probability that $\frac{1}{3} < x < \frac{1}{2}$, i.e. $P\left(\frac{1}{3} < x < \frac{1}{2}\right)$.

Solution: Probability density function $f(x) = cx^2$(1)

Using 2nd property $\int_{-\infty}^{\infty} f(x)dx = 1$

We have

$$\begin{aligned}\int_0^1 cx^2 &= 1 \\ \Rightarrow \frac{1}{3} [cx^3]_0^1 &= 1 \\ \Rightarrow c &= 3\end{aligned}$$

$$P\left(\frac{1}{3} < x < \frac{1}{2}\right) = \int_{1/3}^{1/2} 3x^2 dx = \frac{19}{216}$$

Cumulative distribution function: when value of variate $X \leq x$ then the probability of function $F(x)$ is written as

$$F(x) = \int_{-\infty}^x f(x)dx = P(X \leq x)$$

Then the function $F(x)$ is called the cumulative distribution function or simply the distributive functions.

Properties of cumulative distribution function:

- i.* $F'(x) = f(x) \geq 0$,i.e. $dF(x) = f(x)dx$
- ii.* $F(-\infty) = 0$
- iii.* $F(\infty) = 1$
- iv.* $F(x)$ is a continous function of x on the right.

Q1. the diameter, say X , of an electric cable, is assumed to be a continuous random variable with p.d.f.: $f(x) = 6x(1 - x)$, $0 \leq x \leq 1$

- i. Check that above is p.d.f.
- ii. Obtain an expression for the c.d.f. of X .
- iii. Compute $P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$, and
- iv. Determine the number k such that $P(X < k) = P(X > k)$

Solution:

- i. $\int_0^1 6x(1 - x)dx = 6 \left| \frac{x^2}{2} - \frac{x^3}{3} \right|_0^1 = 1$, it satisfies the first property of probability density function so $f(x)$ is p.d.f.

$$ii. F(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ 3x^2 - 2x^3, & \text{if } 0 < x \leq 1 \\ 1, & \text{if } x > 1 \end{cases}$$

$$iii. P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right) = \frac{P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right)}{P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right)} = \frac{\int_{1/3}^{1/2} 6x(1-x)dx}{\int_{1/3}^{2/3} 6x(1-x)dx} = \frac{11/54}{13/27} = \frac{11}{26}$$

iv. For finding values of k given condition is $P(X < k) = P(X > k)$

$$\Rightarrow \int_0^k 6x(1-x)dx = \int_k^1 6x(1-x)dx$$

$$\Rightarrow 4k^3 - 6k^2 + 1 = 0 \Rightarrow k = \frac{1}{2}, \frac{1 \pm \sqrt{3}}{2} \text{ so value of } k \text{ is } \frac{1}{2} \text{ because it is in given range.}$$

Q1. A random variable X has the following probability mass function:

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

- Find k .
- Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$.
- If $P(X \leq a) > \frac{1}{2}$ find the minimum value of a .
- Determine the distribution function of X .

Q1. A random variable X has the following probability mass function:

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

- Find k .
- Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$.
- If $P(X \leq a) > \frac{1}{2}$ find the minimum value of a .
- Determine the distribution function of X .

Q1. Find the probability distribution of the number of doublets in four throws of a pair of dice.

Q2. The probability distribution of a random variable X is given by:

$$F(x) = \begin{cases} 1 - (1 + x)e^{-x}, & \text{for } x \geq 0 \\ 0, & \text{for } x < 0 \end{cases}$$

find the corresponding density function.

Q3. A random variable x has the density function

$$f(x) = \frac{k}{1+x^2}, -\infty < x < \infty$$

Determine k and the prob. Distribution.

Q4. The diameter (say X) of an electric cable is assumed to be a continuous random variable with pdf $f(x) = 6x(1 - x), 0 \leq x \leq 1$.

(i) Check that function is a pdf. (ii) Obtain an expression for the cdf of X.

(iii) Determine the k such that $P(X < k) = P(X > k)$.

- ✓ Random Variables for one dimension.
- ✓ Probability mass function
- ✓ Probability Density function

Multivariable random variable:

In this lecture, we will extend the idea of single random variable in to the two dimensional variables.

Based on this idea , we will discuss the joint probability distribution, Marginal probability distribution and Conditional probability distribution.

Two dimensional random variables(Multiple random variables):

Let S be the sample space of the random experiment. Let X and Y be two random variables, then the pair (X,Y) is called a bivariate random variable(or two dimensional random variable) if each of X and Y associates a real number with every outcomes.

If the random variables X and Y are both discrete then (X,Y) is called discrete bivariate random variable.

If the random variables X and Y are both continuous then (X,Y) is called continuous bivariate random variable.

For Example: Consider a random experiment of tossing a fair coin thrice. Let the random variable X denote the number of heads and Y denotes the number of tails that occurs in three tosses. Then, X takes the values 0,1,2,3 and Y takes the values 0,1,2,3.

Here (X,Y) is the discrete bivariate random variable.

The range space of X is $R_X = \{0,1,2,3\}$

The range space of Y is $R_Y = \{0,1,2,3\}$

The range space of (X,Y) is $R_{XY} = \{(0,3), (3,0), (1,2), (2,1)\}$

Joint Probability function or distribution:

Let X and Y be two random variables defined on the same sample space.

Case 1: If X and Y are discrete random variables then a probability function

$$P(X = x_i, Y = y_j) = P(x_i, y_j)$$

That yields the probability that X will assume a particular value x_i while at the same time Y assumes a particular value y_j called a joint probability function of X and Y and the following properties:

(i) $P(x_i, y_j) \geq 0$

(ii) $\sum_i \sum_j P(x_i, y_j) = 1.$

Case 2: If X and Y are continuous random variables then the probability density function

$$P\left[x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}, y - \frac{dy}{2} \leq Y \leq y + \frac{dy}{2}\right] = f(x, y)$$

that yields the probability that the point (x, y) lies in the infinitesimal rectangular region of area $dx dy$ is called the joint probability density function of X and Y and following properties:

(i) $f(x, y) \geq 0$ for all (x, y) in the given range.

(ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$ exists and is equal to 1.

(iii) $P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dx dy$

The probability distribution defined in terms of joint probability function or joint probability density function is called a joint probability distribution.

Joint Probability function(CO3)

Example: The joint density function of X and Y is given by

$$P(x, y) = k(2x + y); x = 0, 1, 2; y = 0, 1, 2$$

Find the value of k.

Solution: For joint density function, we have

$$\sum_x \sum_y P(x, y) = 1$$

$$\Rightarrow 0 + 1k + 2k + 2k + 3k + 4k + 4k + 5k + 6k = 1$$

$$\Rightarrow 27k = 1$$

$$\Rightarrow k = \frac{1}{27}.$$

$x \rightarrow$ $y \downarrow$	0	1	2
0	0	2k	4k
1	k	3k	5k
2	2k	4k	6k

Marginal Probability distribution or function:

(i) for two random variables X and Y, the marginal probability distribution of X, denoted by $P(x)$ or P_x , is given as:

For Discrete random variables: $P(x) = \sum_y P(X = x, Y = y)$

For Continuous random variables: $f_X(x) = \int_y f(x, y)dy$

(ii) for two random variables X and Y, the marginal probability distribution of Y, denoted by $P(y)$ or P_y , is given as:

For Discrete random Variables: $P(y) = \sum_x P(X = x, Y = y)$

For Continuous random variables: $f_Y(y) = \int_x f(x, y)dx$

Marginal Probability distribution(CO3)

Example: For the joint probability distribution of two random variables X and Y given below:

Y→ X↓	1	2	3	4	Total
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{10}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{9}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
Total	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	1

Find the marginal probability distribution of X and Y.

Solution: The marginal probability distribution of X is given by

Now,

$$P(1) = \sum_y P(X = 1, Y = y) = P(X = 1, Y = 1) + P(X = 1, Y = 2) \\ + P(X = 1, Y = 3) + P(X = 1, Y = 4)$$

$$\Rightarrow P(1) = \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}$$

Similarly, we can evaluate

$$P(X = 2) = \frac{9}{36}; P(X = 3) = \frac{8}{36}; P(X = 4) = \frac{9}{36}.$$

Now marginal probability distribution of Y is given by

$$P(y) = \sum_x P(X = x, Y = y)$$

$$P(Y = 1) = \frac{11}{36}; P(Y = 2) = \frac{9}{36}; P(Y = 3) = \frac{7}{36}; P(Y = 4) = \frac{9}{36}.$$

Marginal Probability distribution(CO3)

Hence, The marginal distribution of X

X→	1	2	3	4
$P(X)$	$\frac{10}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

The marginal distribution of Y

Y→	1	2	3	4
$P(Y)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

Conditional Probability distribution(CO3)

The conditional Probability distribution for X given Y, denoted by $P(X/Y)$ or $P_{X/Y}$ is defined as :

$$P(X/Y) = \frac{P(X = x, Y = y)}{P(y)} = \frac{P(x, y)}{P(y)} = \frac{\text{Joint density function}}{\text{Marginal Prob. of } Y}$$

Similarly , the conditional Probability distribution for Y given X, denoted by $P(Y/X)$ or $P_{Y/X}$ is defined as :

$$P(Y/X) = \frac{P(X = x, Y = y)}{P(x)} = \frac{P(x, y)}{P(x)} = \frac{\text{Joint density function}}{\text{Marginal Prob. of } X}$$

Conditional Probability distribution(CO3)

Example: The joint density function of X and Y is given by

$$P(x, y) = \frac{(2x+y)}{27}; x = 0,1,2; y = 0,1,2$$

Find the conditional probability distribution of Y given $X = x$.

Solution: The conditional Prob. Distribution of Y when $X = x$ is given:

$$P(Y/X=x) = \frac{P(x,Y)}{P(x)}$$

For $x = 0$:
$$P(Y/X=x) = \frac{P(0,Y)}{P(0)}$$

$$= \frac{0}{3/27} = 0$$

For $x = 1$:
$$P(Y/X=x) = \frac{3}{9}$$

For $x = 2$:
$$P(Y/X=x) = \frac{4}{9}$$

Y→ X↓	0	1	2	Total
0	0	1/27	2/27	3/27
1	2/27	3/27	4/27	9/27
2	4/27	5/27	6/27	15/27
Total	6/27	9/27	12/27	1

Q1. The joint probability distribution of two random variables X and Y is given by:

$$P(X = 0, Y = 1) = \frac{1}{3}; P(X = 1, Y = -1) = \frac{1}{3}; \text{ and } P(X = 1, Y = 1) = \frac{1}{3}.$$

Find (i) Marginal Probability distribution of X and Y(ii) Conditional Probability distribution of X given Y=1.

Q2. The joint probability density function of the two dimensional random variable (X,Y) is of the form:

$$f(x, y) = \begin{cases} e^{-k(x+y)}, & 0 \leq y < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

- (i) Determine the value of k.
- (ii) The marginal Probability density function.
- (iii) The conditional Probability density functions of X and Y.

Q1. The joint probability distribution of X and Y is given by

$$P(x, y) = \frac{x+y}{21}; x = 1, 2, 3; y = 1, 2$$

find the marginal Prob. Distribution of X and Y. Also find the conditional Prob. Distribution of Y given X=3.

Q2. Suppose that two dimensional continuous random variable (X,Y) has joint

pdf given by
$$f(x, y) = \begin{cases} 6x^2y, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Verify that $\int_0^1 \int_0^1 f(x, y) dx dy$

(b) Find $P\left(0 < X < \frac{3}{4}, \frac{1}{3} < Y < 2\right)$.

Q3. Find k so that $f(x, y) = kxy, 1 \leq x \leq y \leq 2$ will be a probability density function.

Q4. A two dimensional random variable (X,Y) have a bivariate distribution given by:

$P(X = x, Y = y) = \frac{x^2 + y}{32}$, for $x = 0, 1, 2, 3$ and $y = 0, 1$. Find the marginal distribution of X and Y.

The concept of independent random variables is very similar to independent events.

Two events A and B are independent if we have

$$P(A, B) = P(A) P(B)$$

$$P(A, B) = P(A \text{ and } B) = P(A \cap B)$$

An independent random variable is a random variable that doesn't have an effect on the other random variables in your experiment. In other words, it doesn't affect the probability of another event happening.

Two random variables are independent if and only if the events related to those random variables are independent events.

The independence between two random variables is also called statistical independence.

1. Two discrete random variable X and Y are said to be independent if

$$P_{XY}(x, y) = P_X(x) \cdot P_Y(y), \quad \forall (x, y) \in R$$

$P_{XY}(x, y)$: Joint probability mass function of two variable

$P_X(x)$: Marginal probability mass function of $X=x$

$P_Y(y)$: Marginal probability mass function of $Y=y$

Two random variables X and Y forming a continuous random vector, are independent if and only if

$$F_{XY}(x, y) = F_X(x) F_Y(y), \quad \forall (x, y) \in R$$

- $F_{XY}(x, y)$: **Joint probability density function of two variable**
- $F_X(x)$: **Marginal probability density function of X= x**
- $F_X(x) = \int_{-\infty}^{\infty} F_{XY}(x, y) dy$
- $F_Y(y)$: **Marginal probability density function of Y= y**
- $F_Y(y) = \int_{-\infty}^{\infty} F_{XY}(x, y) dx$

Example: I toss a coin twice and define X to be the number of heads I observe. Then,

I toss the coin two more times and define Y to be the number of heads that I observe this time.

Find $P(X < 2 \text{ and } Y > 1)$

Solution: since X and Y are result of independent coin tosses.

$P(X < 2 \text{ and } Y > 1) = P(X < 2) P(Y > 1)$, Since X and Y are independent

$$\begin{aligned} &= \{ P_X(X=0) + P_X(X=1) \} P_Y(2) \\ &= (1/4 + 1/2) 1/4 \\ &= 3/16 \end{aligned}$$

Central Limit Theorem (CO3)

The central limit theorem says that the sampling distribution of the mean will always be normally distributed, as long as the sample size is large enough. Regardless of whether the population has a normal, Poisson, binomial, or any other distribution, the sampling distribution of the mean will be normal.

- The central limit theorem (CLT) states that the distribution of sample means approximates a normal distribution as the sample size gets larger, regardless of the population's distribution.
- A sufficiently large sample size can predict the characteristics of a population more accurately.
- Sample sizes equal to or greater than 30 are often considered sufficient for the CLT to hold.
- A key aspect of CLT is that the average of the sample means and standard deviations will equal the population mean and standard deviation.
- CLT is useful in finance and investing when analyzing a large collection of securities to estimate portfolio distributions and traits for returns, risk, and correlation.

- ✓ Multi -variable probability distribution
- ✓ Conditional probability distribution
- ✓ Central limit theorem
- ✓ Statistical independence

Q1. Write central limit theorem.

Q2. Define statistical independence for discrete random variable

Q3. Define statistical independence for continuous random variable

Q1. Four person are chosen at random from a group containing 3 men, 2 women and 4 children then the probability that exactly two of them will be children is

- a. $\frac{9}{21}$
- b. $\frac{10}{21}$
- c. $\frac{6}{21}$
- d. None of these

Q 2. In a certain state, 25 percent of all cars emit excessive amounts of pollutants. If the probability is 0.99 that a car emitting excessive amounts will fail the state's vehicular emission test, and the probability is 0.17 that a car not emitting excessive amounts of pollutants will nevertheless fail the test. What is the probability that a car that fails the test actually emits excessive amounts of pollutants?

- a. 0.60
- b. 0.70
- c. 0.66
- d. None of these

Q3. A can hit a target 3 times in 5 shots, B 2 times in 5 shots and C 3 times in 4 shots. All of them fire one shot each simultaneously at the target. What is the probability that at least two shots hit-

- a. 0.60
- b. 0.62
- c. 0.63
- d. 0.50

Q4. The probability that a civilian can hit a target is $\frac{2}{5}$ and the probability that an army officer can hit the same target is $\frac{3}{5}$ while the civilian can fire 8 shots in the time the army officer fires 10 shots. If they fire together, then what is the probability that army officer shoots the target?

- a. $\frac{6}{11}$
- b. $\frac{5}{11}$
- c. $\frac{3}{11}$
- d. None of these

Q5. The p.d.f $f(x)$ of a continuous random variable x is defined by-

$$f(x) = \begin{cases} \frac{A}{x^3}, & 5 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases} . \text{ Then the value of } A \text{ is-}$$

- a. $\frac{100}{3}$
- b. $\frac{200}{3}$
- c. $\frac{50}{3}$
- d. None of these

Glossary Questions (CO3)

Q: Please select the correct option from the given Glossary:

(i) 1 (ii) $\frac{1}{3}$ (iii) (0,1,2) (iv) $\frac{19}{216}$ (v) $\frac{P(x,y)}{P(y)}$.

(a) The value of k _____ so that function $f(x) = ke^{-x/3}$, $0 \leq x < \infty$ will be a probability density function.

(b) The random variable takes the value _____ from the number of aces in a draw of two cards from a well shuffled deck of 52 cards.

(c) The value of $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy$ is equal to _____. If $f(x,y)$ is a probability density function.

(d) The formula of the Conditional probability distribution X given Y is _____.

(e) If the pdf of $f(x) = 3x^2$; $0 \leq x \leq 1$, then the $P\left(\frac{1}{3} < x < \frac{1}{2}\right)$ is _____.

Glossary Questions (CO3)

Q: Please select the correct option from the given Glossary:

(i) 1 (ii) $\frac{1}{3}$ (iii) (0,1,2) (iv) $\frac{19}{216}$ (v) $\frac{P(x,y)}{P(y)}$.

(a) The value of k _____ so that function $f(x) = ke^{-x/3}$, $0 \leq x < \infty$ will be a probability density function.

(b) The random variable takes the value _____ from the number of aces in a draw of two cards from a well shuffled deck of 52 cards.

(c) The value of $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy$ is equal to _____. If $f(x,y)$ is a probability density function.

(d) The formula of the Conditional probability distribution X given Y is _____.

(e) If the pdf of $f(x) = 3x^2$; $0 \leq x \leq 1$, then the $P\left(\frac{1}{3} < x < \frac{1}{2}\right)$ is _____.

Q1. Let X represents the difference between the number of heads and the number of tails obtained when a coin is tossed 6 times. What are possible values of X ?

Q2. Find the probability distribution of

- (i) number of heads in two tosses of a coin
- (ii) number of tails in the simultaneous tosses of three coins
- (iii) number of heads in four tosses of a coin

Q3 Find the probability distribution of the number of success in two tosses of die, where a success is defined as

- (i) number greater than 4

Expected University Questions (CO3)

Q4. A random variable X has the following probability mass function:

x	0	1	2
$p(x)$	$3c^3$	$4c-10c^2$	$5c - 1$

I. Find c .

II. Evaluate $(X < 2)$, $(2X + 3 \geq 5)$ and $(1 < X \leq 2)$.

III. If $(X \leq a) < 1$, find the maximum value of a .

Q5. If X and Y are two random variables having joint density function:

$$f(x, y) = \begin{cases} 6(1 - x - y) & 0 \leq x < 2, 2 \leq y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find marginal density function of X and Y . Hence examine if X and Y are independent

[First Sessional Set-1 \(CSE,IT,CS,ECE,IOT\).docx](#)

[Second Sessional Set-2 \(CSE,IT,CS,ECE,IOT\).docx](#)

[Maths IV PUT.docx](#)

End Semester Question Paper

Printed Page:- 06

Subject Code:- BAS0303

Roll. No:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY, GREATER NOIDA
(An Autonomous Institute Affiliated to AKTU, Lucknow)

B.Tech

SEM: III - THEORY EXAMINATION (2024- 2025)

Subject: Statistics & Probability

Time: 3 Hours

Max. Marks: 100

General Instructions:

IMP: Verify that you have received the question paper with the correct course, code, branch etc.

1. This Question paper comprises of **three Sections -A, B, & C**. It consists of Multiple Choice Questions (MCQ's) & Subjective type questions.
2. Maximum marks for each question are indicated on right -hand side of each question.
3. Illustrate your answers with neat sketches wherever necessary.
4. Assume suitable data if necessary.
5. Preferably, write the answers in sequential order.
6. No sheet should be left blank. Any written material after a blank sheet will not be evaluated/checked.

SECTION-A

20

1. Attempt all parts:-

1-a. Wheat crops badly damaged on account of rains is: (CO1, K2)

1

- (a) Cyclical movement
- (b) Random movement
- (c) Secular trend
- (d) Seasonal movement

End Semester Question Paper

- 1-b. Let the average of three numbers be 16. If two of the numbers are 8 and 12, then the remaining number is..... (CO1, K3) 1
- (a) 28
(b) 18
(c) 12
(d) 30
- 1-c. One card is drawn from a standard pack of 52 plying cards. Find the probability that it is either a king or a queen. (CO2, K3) 1
- (a) $\frac{1}{13}$
(b) $\frac{2}{13}$
(c) $\frac{3}{13}$
(d) None of these
-
- 1-d. If two events A and B are mutually exclusive, then the probability $P(A \cap B)$ is: (CO2, K2) 1
- (a) 0
(b) 1
(c) $P(A).P(B)$
(d) None of these

End Semester Question Paper

- | | | |
|------|---|---|
| 1-e. | <p>In binomial distribution probability of success in each trial remains _____.(CO3, K1)</p> <p>(a) 0
(b) 1
(c) Constant
(d) Not defined</p> | 1 |
| 1-f. | <p>In Normal Distribution, Mean deviation about mean is (CO3, K1)</p> <p>(a) σ
(b) $2\sigma/5$
(c) $4\sigma/5$
(d) $6\sigma/7$</p> | 1 |
| 1-g. | <p>The area of critical region depends on the size of.... (CO4, K1)</p> <p>(a) Type I error
(b) Type II error
(c) Test statistics
(d) Sample</p> | 1 |
| 1-h. | <p>In conducting one way analysis of variance --- test statistics would be used. (CO4,K1)</p> <p>(a) Z
(b) T
(c) Chi-Square
(d) F</p> | 1 |

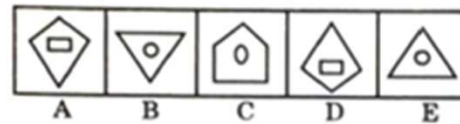
End Semester Question Paper

1-i. If $A = \{1,4,6\}$, $B = \{3,6\}$ and $C = \{3,4,6\}$ then $A \cap (B \cap C)$ is.... (CO5,K3) 1

- (a) $\{3,4,5,6\}$
- (b) $\{4,6\}$
- (c) $\{1,4,6\}$
- (d) $\{6\}$

1-j. Select a figure from amongst the answer figure which will continue the same series as established by the five-figure problem. (CO5,K2) 1

Problem Figure



Answer Figure



- (a) 1
- (b) 2
- (c) 3
- (d) 5

End Semester Question Paper

2. Attempt all parts:-

- | | | |
|------|--|---|
| 2.a. | Prove that if two variables are independent, their correlation is zero but vice versa is not true. (CO1,K2) | 2 |
| 2.b. | Find the probability of getting 53 Sundays in a leap year. (CO2,K3) | 2 |
| 2.c. | A Binomial random variable X satisfies the relation $9P(X=4) = P(X=2)$, When $n=6$. Find value of $P(X=1)$. (CO3, K3) | 2 |
| 2.d. | Define an estimator in statistics. (CO4, K1) | 2 |
| 2.e. | Check whether the function $f: N \rightarrow N$ is defined by $f(x) = x^2 + 12$ is one-one or not. (CO5, K3) | 2 |

SECTION-B

30

3. Answer any five of the following:-

- | | | |
|------|--|---|
| 3-a. | The mean and standard deviation of the marks of 100 candidates was found to be 40 and 5.1, respectively. Later, it was discovered that a score of 40 was wrongly read as 50. Find out the correct mean and standard deviation respectively. (CO1,K3) | 6 |
| 3-b. | Calculate Spearman's rank correlation coefficient from the following data: (CO1, K3) | 6 |

End Semester Question Paper

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

- 3-c. A random variable X has the following distribution, Find the value of K. Also find Mean and variance (CO2, K3) 6

x	-2	-1	0	1	2	3
P(x)	0.1	K	0.2	2K	0.3	K

- 3-d. A bag X contains 2 white and 3 red balls and another bag Y contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag Y. (CO2, K3) 6

- 3.e. If 10% of the bolts produced by a machine are defective, determine the probability that out of 10 bolts chosen at random 6
- 1
 - None
 - at most 2 bolts will be defective. (CO3,K3)

- 3.f. A sample of 20 items has mean 42 units and S.D. 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units. (If the tabular value at 5% LOS for 19 d. f. is 2.09). (CO4, K3) 6

- 3.g. How many different words can be formed using all the letters of the word ALLAHABAD 6

- When the vowels occupy the even position.
- Both L do not occur together. (CO5,K3)

End Semester Question Paper

SECTION-C

50

4. Answer any one of the following:-

4-a. In a partially destroyed laboratory record of analysis of a correlation data, the 10

following results only are legible: (CO1,K3)

Variance of $x = 9$; Regression equations: $8x - 10y + 66 = 0$, $40x - 18y = 214$.

What were (a) the mean values of x and y (b) the standard deviation of y (c) the coefficient correlation between x and y .

4-b. Fit a second degree parabola by the method of least squares to the following data 10
.(CO1, K3)

X	0	1	2	3	4
Y	1	4	10	17	30

5. Answer any one of the following:-

5-a. State and prove Bayes theorem. 10

In a Neighbourhood, 90% children were falling sick due flu and 10% due to measles and no other disease. The probability of observing rashes for measles is 0.95 and for flu is 0.08. If a child develops rashes, find the child's probability of having flu. (CO2, K3)

End Semester Question Paper

5-b. Let the two dimensional continuous random variable (X,Y) has joint PDF given by 10

$$f(x,y) = \begin{cases} 6x^2y, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find (i) $P(0 < x < 3/4, 1/3 < y < 2)$ (ii) $P(x + y < 1)$. (CO2, K3)

6. Answer any one of the following:-

6-a. A manufacturer of envelopes knows that the weight of the envelopes is normally 10

distributed with mean 1.9gm and variance 0.01 square gm. Find how many

envelopes weighing (i) 2gm or more

(ii) 2.1gm or more, can be expected in a given packet of 1000 envelopes?

Given that the area under the standard curve

between $z = 0$ and $z = 1$ is 0.3413,

between $z = 0$ and $z = 2$ is 0.4772. (CO3, K3)

6-b. 10

Four coins were tossed 200 times. The number of tosses showing 0,1,2,3 and 4 heads were found to be as under.

Fit a binomial distribution to these observed results. Find the expected frequencies. (CO3, K3)

No. of Heads	0	1	2	3	4
No. of Tosses	15	35	90	40	20

End Semester Question Paper

7. Answer any one of the following:-

7-a.

10

To test of significance of the variations of the retail prices in the commodity in three principal cities: Mumbai, Bangalore and Chennai. The four shops were chosen at random in each city and prices observed in INR were as follows:

Mumbai	16	8	12	14
Bangalore	14	10	10	6
Chennai	4	10	8	8

Do the data indicate that the prices in the three cities are significantly different? Given that the tabular value of F is 4.26 5% LOS with d.f. is (2,9). (CO4,K3)

7-b. From the following data, find whether hair color and gender are associated.

10

End Semester Question Paper

<i>Gender</i> ↓	<i>Colour</i>					
	Fair	Red	Medium	Dark	Black	Total
Boys	529	849	504	119	36	2100
Girls	544	677	451	97	14	1783
Total	1136	1526	955	216	50	3883

Given that the tabular value of χ^2 is 9.488 at 5%LOS with d.f. 4. (CO4, K3)

8. Answer any one of the following:-

8-a. Solve the following: (CO5,K3)

10

1. What is the sum of all five-digit numbers formed by 2, 3, 4, 5, 6 without repetition?
2. What is the sum of all five-digit numbers formed by 2, 3, 4, 5, 6 with repetition?

End Semester Question Paper

8-b.

Study the following table and answer the questions based on it
Expenditures of a Company (in Lakh Rupees) per Annum Over the given Years.

10

Year	Item of Expenditure				
	Salary	Fuel and Transport	Bonus	Interest on Loans	Taxes
1998	288	98	3	23.4	83
1999	342	112	2.52	32.5	108
2000	324	101	3.84	41.6	74
2001	336	133	3.68	36.4	88
2002	420	142	3.96	49.4	98

- Find the average amount of interest per year which the company had to pay during this period?
- The total amount of bonus paid by the company during the given period is approximately what percent of the total amount of salary paid during this period?
- Total expenditure on all these items in 1998 was approximately what percent of the total expenditure in 2002?
- The total expenditure of the company over these items during the year 2000 is?
- The ratio between the total expenditure on Taxes for all the years and the total expenditure on Fuel and Transport for all the years respectively is approximately? (CO5,K3)

Expected Questions for University Exam(CO1)

Q1: An incomplete distribution of families according to their expenditure per week is given below. The median and mode for the distribution are Rs 25 and Rs 24 respectively. Calculate the missing frequencies.

Expenditure	0-10	10-20	20-30	30-40	40-50
No. of families	14	?	27	?	15

Q2: The first four moments of a distribution about 2 are 1,2.5,5.5 and 16 respectively. Calculate the four moments about mean and about the origin.

Expected Questions for University Exam(CO1)

Q3: Calculate the mean and standard deviation for the following data:

Size of item	6	7	8	9	10	11	12
Frequency	3	6	9	13	8	5	4

Q4: The first four moments of the distribution about the value 4 of the variable are -1.5, 17, -30, and 108. Find the moments about mean, β_1 and β_2 .

Text Books

- N. P. Bali: A Textbook of Engineering Mathematics-IV, University Science Press.
- H. K. Dass: Introduction to Engineering Mathematics, S. Chand.
- S. Ross: A First Course in Probability, 6th Ed., Pearson Education India, 2002.
- W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968.

Reference Books

- P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003(Reprint).
- R.K. Jain and S.R.K. Iyenger: Advance Engineering Mathematics; Narosa Publishing House, New Delhi.
- J.N. Kapur: Mathematical Statistics; S. Chand & Sons Company Limited, New Delhi.
- D.N.Elhance, V. Elhance & B.M. Aggarwal: Fundamentals of Statistics; Kitab Mahal Distributers, New Delhi.

Thank You